Strategy for Constructing a Monthly Capital (K) Time Series for the Entire Haifa Port

1. Objective and Scope

Goal: Build a defensible, documented **monthly capital stock series $K_n^{\text{Tydshytyl}}$** for the **Haifa container port as a whole**, so we can compute **monthly K/L and LP** at the **port level**, not just for a single terminal.

The Haifa container port is served by multiple owners/operators:

- **Haifa Port Company (HPC, legacy + Adani)** – legacy container terminal and other activities.
- **SIPG Bayport Terminal Co. Ltd. (Bayport)** – new entrant container terminal.
- **Israel Ports Development & Assets Co. (IPC / Hani)** – landlord owning much of the infrastructure, especially for Bayport.

This strategy explains how to assemble a **monthly, real K series** for the **entire Haifa container system** by combining:

1. A high-quality **firm-level K series for HPC** (already built).
2. A project-level / firm-level **K series for SIPG Bayport**, built from SIPG group reports and (ideally) stand-alone FS.
3. A project-level **infrastructure K series for IPC's Haifa container assets**, especially Bayport.

We target a hierarchy of tracks:

- **Track B (HPC)** – firm-level, Note-8-based K for Haifa Port Company (legacy/Adani).
- **Track C (SIPG + IPC Bayport projects)** – project-level K for entrant and landlord capital.
- **Track D (Haifa container sector)** – combined port-wide K used for K/L and LP at the port level.

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2. Conceptual K for the Haifa Container Port

2.1 Decomposition

Define **Haifa container-sector K** as:

$$K_t^{\text{Haifa, cont}} = K_t^{\text{HPC, cont}} + K_t^{\text{SIPG}} + K_t^{\text{IPC, Haifa-cont}}$$

Where:

- **K^{HPC, cont}_t** = capital stock directly owned by Haifa Port Company and used in container operations (quays, cranes, yards, buildings, equipment) at time t.
- **K^{SIPG}_t** = capital stock owned by SIPG Bayport Terminal Co. Ltd. (STS cranes, yard cranes, IT/TOS, other superstructure) used in Haifa Bayport container operations.

- **K^{IPC, Haifa-cont}_t** = container-relevant infrastructure at Haifa owned by IPC (breakwaters, reclaimed land, quay walls, dredging, some utilities and access works).

All components will be expressed in **constant NIS** (or a chosen base currency) and in **net, not gross, capital**, consistent with the OECD-style PIM you are already using.

2.2 Time Frequency and Alignment

We ultimately want **monthly K_t** for alignment with:

- Monthly L (hours) and LP (TEU per hour), and
- Monthly event-time indicators around Bayport opening and privatization.

But data will arrive at various frequencies:

- Annual FS for HPC, SIPG, IPC.
- Project-level annual or multi-year cost envelopes.
- Occasional mid-year balance sheets (e.g., 2021-06 Bayport SPV).

Approach:

1. Build **annual K series** via PIM for each component.
2. Disaggregate to **monthly K_t** using:
3. Step changes in commissioning months for big projects.
4. Smooth interpolation for background investment and depreciation.

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3. HPC Component – Track B (Anchor)

3.1 Current Status

You already have a detailed **HPC K series** constructed from:

- Annual HPC financial statements (Note 8 PPE, depreciation, capex, disposals).
- Cash-flow statements and depreciation notes.
- A project-level table (`Haifa_big_projects.csv`) identifying lumpy investments and disposals.
- A PIM implementation with:
- Real deflators (capital_deflator.csv).
- Asset-class-specific lives and geometric depreciation.
- Hybrid disaggregation to monthly K with big-project jumps.

This forms **Track B: Haifa legacy firm-level K**.

3.2 Restricting to Container Operations

To get **K^{HPC, cont}_t**, refine your existing HPC K as follows:

1. **Tag big projects**:

2. In `Haifa_big_projects.csv`, classify each project as:
    - `container_core` (container quays, STS cranes, RTG/RMG, container yards),
    - `container_shared` (shared infrastructure used partly by containers),
    - `non_container` (bulk, cruise, general cargo, admin buildings unrelated to containers).
3. **Allocate investment by class**:
4. Include 100% of `container_core` investment.
5. Include an appropriate share (e.g., 50%) of `container_shared` based on engineering judgment or throughput shares.
6. Exclude `non_container` projects from the container K series.
7. **Run a parallel PIM** with only the selected projects and the corresponding portions of annual additions and stocks.

Result: a **monthly container-specific K series for HPC**, $K_t^{HPC,cont}$, which becomes the **anchor component** of the Haifa port K.

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4. SIPG Bayport Component – Track C, Part 1 (Using Existing Data)

4.1 What We Already Have

From the SIPG group reports and related filings you have already downloaded, you can extract for the **"Israel Haifa Bayport Port Engineering Project"**:

- **Total project budget** in RMB.
- Annual **Construction in Progress (CIP)** balances for the project:
- Opening balance.
- Additions during the year.
- Transfers to fixed assets and intangible assets.
- Ending balance.
- Cumulative investment share of budget.
- The year when the project reaches **100% physical completion** and CIP is fully transferred to fixed assets.

Additionally, there is at least one **Bayport-SPV stand-alone balance sheet** (mid-2021) that shows total assets and fixed assets for SIPG Bayport Terminal Co. Ltd.

4.2 Building an Initial Project-Level PIM for Bayport (Approximate K^{SIPG}_t)

Step 1: Extract annual investment flows I^{SIPG}_y

From each year's SIPG consolidated FS:

1. Construct a table with rows for each fiscal year y and columns:
2. Project name (Haifa Bayport).
3. CIP opening balance (CIP_y0).
4. Additions to CIP during year (I_y).
5. Transfers to fixed assets / intangibles.

6. CIP closing balance (CIP_y1).
7. Cumulative investment ratio.
8. Compute **annual nominal investment** for Bayport as:
9. I_y = additions to CIP during year y (plus any interest capitalised, if you want to include it as part of K).

Step 2: Currency and deflation

1. Decide on a **base currency**:
2. Option A: Keep the PIM in RMB and only convert to NIS or USD at the end.
3. Option B: Convert each year's I_y into NIS at the average FX rate for that year, then deflate with your **Israeli capital-goods deflator** for comparability with HPC.
4. Apply your capital deflator to get **real I_y** and real CIP_y.

Step 3: PIM during construction

1. During the construction years (before commissioning), assume **no depreciation** of CIP (or a very low rate); CIP is just accumulated real investment.
2. Your PIM recursion is:
3. K^{SIPG}_0 = 0.
4. K^{SIPG}*y = (1 - \delta* + I_y.}) K^{SIPG}_{y-1
5. With \delta_{pre} ≈ 0 until commissioning.

Step 4: Commissioning and post-commissioning depreciation

1. Identify the year (and ideally month) when Bayport becomes operational (e.g., 2021-09) and when CIP is fully transferred to fixed assets.
2. From that point onward, apply **standard port-equipment depreciation** (your existing asset-class lives):
3. For STS cranes, yard cranes, etc. use the same life assumptions as for HPC.
4. For Bayport's share of civil works still held by SIPG (if any), use longer civil-works lives.
5. Continue the PIM with \delta_{post} > 0:
6. K^{SIPG}*y = (1 - \delta* + I_y (if any post-commissioning investment occurs).}) K^{SIPG}_{y-1

Step 5: Monthly disaggregation

1. Assign each year's I_y to months according to a rule:
2. **Simple option:** allocate I_y uniformly across the 12 months of year y.
3. **Better option:** concentrate I_y in a few key months based on project milestones (e.g., a large share in the commissioning year around go-live).

4. Interpolate K^{SIPG}_t monthly between annual K^{SIPG}_y values, adding monthly investment and depreciation.

This yields an **approximate monthly K^{SIPG}_t** that reflects the scale and timing of Bayport's capital buildout, even before you obtain stand-alone Bayport FS.

4.3 Using the 2021 Stand-alone Balance Sheet as a Calibration Point

To improve this approximate series:

1. Compare your PIM-based K^{SIPG}_{2021-06} (or 2021-12) with the **net PPE + CIP** values reported in the Bayport-SPV balance sheet.
2. If necessary, scale or adjust your depreciation assumptions so that the PIM approximate K is close to the reported level.

This calibration step helps ensure your Bayport K series is **anchored to at least one true accounting observation**.

4.4 Limitations and Labeling

Because this approach relies on **project-level consolidated CIP** rather than full Bayport-SPV Note 8, label this component as:

- **Track C1: Bayport K derived from SIPG group project disclosures.**

Be explicit in the thesis that Track C1 is approximate and primarily used for **port-level descriptive analysis** and **port-level K/L**, not as the main input into terminal-level K/L regressions.

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5. SIPG Bayport Component – Track C, Part 2 (Getting Better Data)

To upgrade from an approximate project-level K to a **full Bayport terminal K series analogous to HPC**, pursue these routes in parallel.

5.1 Route A – Israeli Corporations Authority / Local Providers

Goal: Obtain **audited annual financial statements** for SIPG Bayport Terminal Co. Ltd. (company no. 515246833) filed in Israel.

Steps:

1. **Identify the company**:
2. Name in Hebrew: אס.איי.פי.ג'י. טרמינל בייפורט חברה בע"מ.

3. Company number: 515246833.

4. **Contact Israeli business-information providers** (CheckID, KYC, BDI, D&B Israel) and explicitly request:

5. The full set of **audited annual FS (דוחות כספיים מבוקרים)** and notes for this company for all available years.

6. Specify that you need:

    - Balance sheet.
    - Income statement.
    - Cash-flow statement.
    - Note on **PPE (רכוש קבוע)**, including opening balance, additions, disposals, depreciation.

7. Alternatively, **engage a local CPA or corporate lawyer** to:

8. Log into the **Rasham HaHavarot** (Corporations Authority) system.
9. Purchase all scanned documents for company 515246833.
10. Identify and forward the FS PDFs.

Once obtained, build a `sipg_bayport_financials_raw.tsv` paralleling `haifa_financials_raw.tsv` and run a full **Path-B PIM**.

5.2 Route B – Chinese Regulatory Filings and Corporate Databases

Because SIPG Bayport is ultimately owned by a Shanghai-listed group, its SPV-level FS may also be filed with **Chinese regulators** (MOF, regional tax bureaus) or captured in corporate databases.

Steps:

1. Use Chinese corporate-information platforms (Qichacha, Tianyancha, etc.) to search for:
2. SIPG BAYPORT TERMINAL CO. LTD.

3. Possible Chinese transliteration or Chinese name for the entity.

4. Retrieve and download any available **annual financial statements** for the SPV:

5. 资产负债表 (balance sheet).
6. 利润表 (income statement).
7. 现金流量表 (cash-flow statement).

8. Any notes detailing PPE and CIP.

9. Translate key tables and incorporate them into `sipg_bayport_financials_raw.tsv`.

5.3 Route C – Direct Contact with SIPG / Bayport

Complement the document-based routes with a **direct request**:

1. Email **SIPG Investor Relations** and/or Bayport management, explaining:
2. You are an academic researcher at UC Berkeley.
3. You are studying the impact of port reforms on productivity and capital deepening.
4. You already have detailed K-series for Haifa legacy.
5. You are seeking **terminal-level PPE and depreciation data** for the Bayport SPV for research purposes.
6. Request either:
7. Full audited FS for Bayport SPV for 2017–2024, or
8. A summary panel of net PPE, annual additions, disposals, and depreciation at year-end.

Any positive response here can dramatically upgrade Track C from approximate to almost Path-B quality.

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6. IPC Haifa Infrastructure Component – Track C, Part 3

6.1 What IPC Controls

IPC (Israel Ports Development & Assets Co.) is the **landlord** for Israel's ports. It owns:

- **Bayport civil works**: reclaimed land, quay walls, breakwaters, dredged approaches, container-yard paving.
- Large shared infrastructure at Haifa: access roads, internal rail, some utilities and common-use structures.

This is crucial because a significant fraction of **Bayport's physical capital** is on IPC's balance sheet rather than SIPG's or HPC's.

6.2 Data Sources for IPC

1. **IPC Annual and Interim Financial Statements**:
2. Available via IPC's website and gov.il.
3. Include:
    - Total net PPE (often broken down by asset class).
    - Construction-in-progress, with project descriptions (including Haifa Bayport and Ashdod Southport).
4. **Project-level Reports and Prospectuses**:

5. Government reports and project descriptions for the Bayport civil-works project, often quoting total costs and milestones.

6.3 Building an IPC Project Table for Haifa

Construct a `ipc_haifa_infra_projects.csv` with one row per major project element, including:

- Project ID.
- Owner: IPC.
- Location: Haifa Bayport / Haifa existing container quays.
- Description: breakwater, dredging, quay construction, land reclamation, yard paving, etc.
- Nominal cost (best estimate) and currency.
- Start year and commissioning year (and month if available).
- Asset class and life (e.g., breakwaters 50 years, quays 40 years, yards 30 years).
- Sources and notes.

6.4 IPC Haifa Infra PIM

Using IPC FS and the project table:

1. Identify the **annual nominal investment flows** related to Haifa Bayport infra.

2. If FS don't cleanly separate Bayport, approximate using project-cost envelopes and IPC's CIP movements.

3. Convert to real I^{IPC, Haifa-cont}_y using the capital deflator.

4. Run a PIM with **long asset lives** appropriate for port infrastructure.

5. Disaggregate to **monthly K^{IPC, Haifa-cont}_t** with large step changes when major elements (breakwater, berths) are commissioned.

6.5 Pre-Bayport IPC Infrastructure

For the pre-Bayport period:

- Either keep IPC infra implicit (folded into HPC K where relevant), or
- Construct a simple background K^{IPC, Haifa-pre}_t representing legacy Haifa infrastructure:
- Use early IPC FS to set an initial infra K level.
- Assume slow growth/depreciation until Bayport construction begins.

This component matters most for capturing **Bayport's infra jump**; the pre-Bayport baseline can be relatively coarse.

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7. Combining Components into Monthly Haifa-Port K

7.1 Annual Combination

Once annual PIMs are built for each component, construct an annual **Haifa container K** as:

$$K_y^{\text{Haifa, cont}} = K_y^{\text{HPC, cont}} + K_y^{\text{SIPG}} + K_y^{\text{IPC, Haifa-cont}}.$$

Ensure all components:

- Are in the **same currency and price base**.
- Use **consistent net-of-depreciation definitions**.

7.2 Monthly Disaggregation and Alignment

1. For each component, have a monthly K series K_t derived from:
2. Big-project step changes at commissioning months.
3. Smooth interpolation of residual investment/depreciation.
4. Sum component-wise to get:

$$K_t^{\text{Haifa, cont}} = K_t^{\text{HPC, cont}} + K_t^{\text{SIPG}} + K_t^{\text{IPC, Haifa-cont}}$$

1. Align this K^{Haifa, cont}_t with:
2. Monthly L^{Haifa}_t (sum of HPC + Bayport labor hours once Bayport opens).
3. Monthly TEU^{Haifa}_t (sum of container throughput across both terminals).
4. Compute:
5. **Port-level K/L**: K^{Haifa, cont}_t / L^{Haifa}_t.
6. **Port-level LP**: TEU^{Haifa}_t / L^{Haifa}_t.

These become the variables for **port-level descriptive analysis** and, if desired, port-level versions of Models 1B and 2.

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8. Multiple Paths and Fallbacks for Each Missing Piece

8.1 SIPG Bayport K

- **Path 1 (now):** Use SIPG consolidated CIP tables and budget to build **Track C1** project-level K via PIM. This is feasible immediately with existing PDFs.

- **Path 2 (better):** Acquire Bayport SPV FS from Israeli providers and build **Track C2** Path-B K (same quality as HPC). Requires money and/or professional help.
- **Path 3 (complementary):** Acquire SPV FS from Chinese corporate databases and triangulate with Path 2.
- **Path 4 (long-shot but high value):** Direct cooperation with SIPG/Bayport management or IR to obtain a curated K panel.

8.2 IPC Haifa Infra K

- **Path A (FS-based):** Use IPC FS CIP notes to estimate Bayport infra investment by year and run a PIM.
- **Path B (project-envelope-based):** Use total project cost and milestone dates from government/engineering reports and simulate a plausible investment schedule.
- **Path C (hybrid):** Anchor the simulation on any available IPC CIP numbers and adjust the envelope to match IPC's total infra K growth.

8.3 General Fallbacks

Even if you cannot reach Path C2/B2 (full Bayport/IPC micro FS), you still have:

- A **high-quality HPC K series** (Track B).
- A **well-documented approximate Bayport K** from SIPG project data (Track C1).
- A **reasonable IPC infra K** from project envelopes and FS (Track C).

Together, these give you a robust **port-level K path** that captures:

- The **magnitude and timing** of the capital deepening at Haifa.
- The relative contributions of legacy, entrant, and landlord capital.

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9. How This Supports Your Econometric Design

9.1 Terminal-Level Models

With this strategy, you maintain:

- **Terminal-level LP event studies (Model 1A)** for all terminals.
- **Terminal-level K/L and mediation** for Haifa legacy (and later Bayport, if you upgrade its K to Path B).

9.2 Port-Level Extensions

Using Track D K^{Haifa, cont}_t, you can add:

- **Port-level K/L event studies** (reform → K/L for Haifa as a whole).
- A **port-level LP – K/L regression**, complementing terminal-level Model 2.
- A **port-level mediation narrative**: how much of the port-level LP change can plausibly be described as driven by capital deepening.

9.3 Transparency and Credibility

At every step, document:

- Which component is Path B (FS-based, micro).
- Which is Track C1/C2 (project-level approximations).
- What assumptions you make on asset lives, FX, and deflation.

Make it clear in the thesis that:

- **HPC K** is your gold-standard K.
- **SIPG and IPC K** are carefully calibrated approximations, sufficient to capture the main capital deepening pulse at Haifa and to support port-level analysis.

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10. Summary

To get a **monthly K series for the entire Haifa container port**, you will:

1. Use your existing **HPC Path-B K** as the anchor and refine it to container-specific K.
2. Build an approximate **Bayport K** from SIPG project-level CIP data (Track C1), and pursue better Bayport FS via Israeli and Chinese channels (Track C2).
3. Build an **IPC Haifa infra K** from IPC FS and project envelopes, focusing on Bayport infra and major Haifa container works.
4. Combine these into a **monthly Haifa-port container K** and align with port-level L and TEU to obtain K/L and LP for the entire Haifa container system.

This strategy respects your thesis' emphasis on **Haifa as a system**, while acknowledging data asymmetries between legacy and entrant operators and providing multiple upgrade paths as additional FS become available.